On Solving the Sparse Matrix Compression Problem

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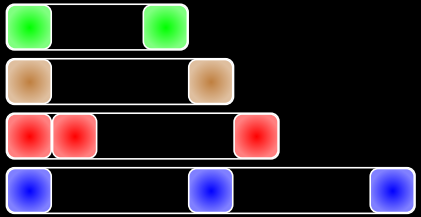
game

slides

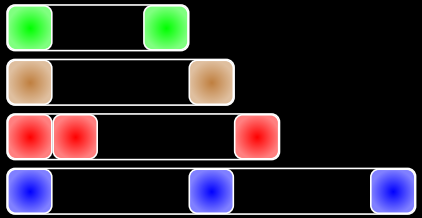


Shonan Meeting No. 240

problem setting

given

- n 1-dimensional tiles
- a tile consists of blocks and gaps



task

- combine all n tiles to a single tile, called placement
- can fill gaps but blocks must not overlap
- goal: construct shortest placement

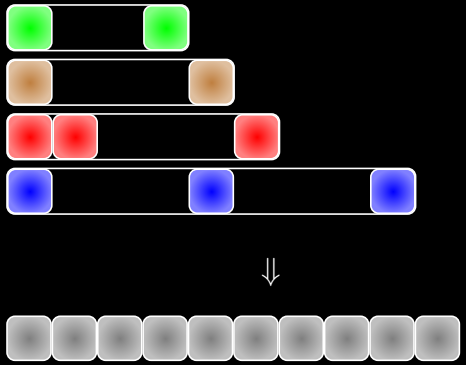
problem setting

Lemma

a computed placement with no gaps is a solution

Proof.

because blocks cannot overlap □



decision problems

MINLENGTH can you combine all tiles to a placement of length k?

MINMAXSHIFT if the first block of each tile is on the first column, can you form a placement with a maximum shift to the right of at most k?

turns out that MINMAXSHIFT has already been studied under the name Sparse Matrix Compression (SMC) problem

- Garey,Johnson'79 showed that SMC is $\mathcal{NP}$-hard for $k \geq 2$
- Bannai+'24 showed that both problems are $\mathcal{NP}$-hard even for widths in $\Omega(\lg n)$

If all tiles have the same length, then MINLENGTH = MINMAXSHIFT

Problem (SMC, [Garey,Johnson'79, Chapter A4.2])

given:
- $n \times \ell$ matrix $A[1..n][1..\ell]$ with n rows and ℓ columns and entries $A[i][j] \in \{0,1\}$ for all $i \in [1..n]$, $j \in [1..\ell]$
- integer $k \in [0..\ell \cdot (n-1)]$

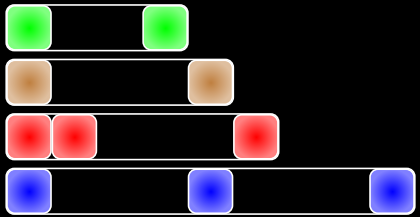
goal: check whether the following two can exist:

- an integer array $C[1..\ell + k]$ with $C[i] \in [0..n]$ for every $i \in [1..\ell + k]$, and
- a shift function $s : [1..n] \to [0..k]$ such that $A[i][j] = 1 \Leftrightarrow C[s(i) + j] = i \;\forall\; i \in [1..n],\; \forall\; j \in [1..\ell]$
- assume $A[0][j] = 0 \;\forall\; j$ to allow setting $C[i] = 0$ for some i, modelling that this entry is unassigned

applications:

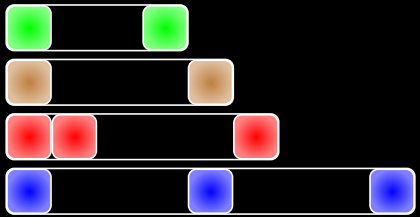
- matrix compression [Ziegler'77]
- search trie implementations [Tarjan,Yao'79]
- compilers [Aho+'86]
- Bloom filters [Chang,Wu'91]

from tiles to matrix



$$A = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

from tiles to matrix



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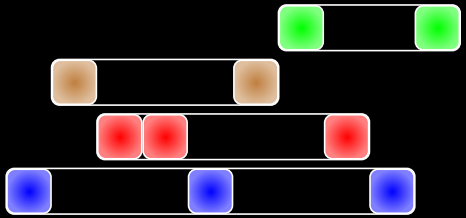
solution:

- $s = [6, 1, 2, 0]$
- $C = [4, 2, 3, 3, 4, 2, 1, 3, 4, 1]$

$$B = \begin{pmatrix} & & & & & & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & & & & & \\ & & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & & & & \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & & & & & & \end{pmatrix}$$

from tiles to matrix



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constant-time access query

sparse matrix representation of A

- $s = [6, 1, 2, 0]$
- $C = [4, 2, 3, 3, 4, 2, 1, 3, 4, 1]$

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

- access query $A[r][c]$ (row r, column c)
- $s[r]$ is the shift of row r
- so $A[r][c] = 1$ if and only if $C[c + s[r]] = r$

Example

- access query $A[3][2]$ (3rd row, 2nd column)
- $s[3] = 2$, so 3rd row shifted by 2
- $C[2 + s[3]] = C[4] = 3$, so $A[3][2] = 1$

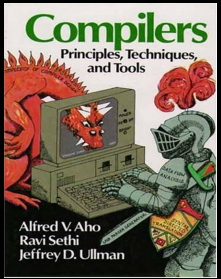
approximation algorithm

Ziegler'77: greedy algorithm: first fits first

- place first tile at first position
- for each subsequent tile: put it at the leftmost fitting position
- repeat

used in the classic textbook "Compilers: Principles, Techniques, and Tools", Section 3.9.8

While we may not be able to choose *base* values so that no *next-check* entries remain unused, experience has shown that the simple strategy of assigning *base* values to states in turn, and assigning each *base*[*s*] value the lowest integer so that the special entries for state *s* are not previously occupied utilizes little more space than the minimum possible.



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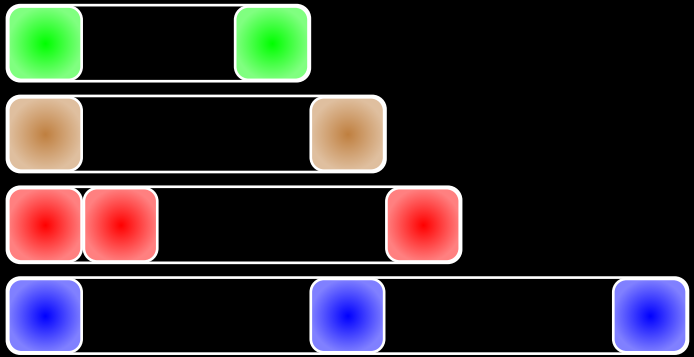

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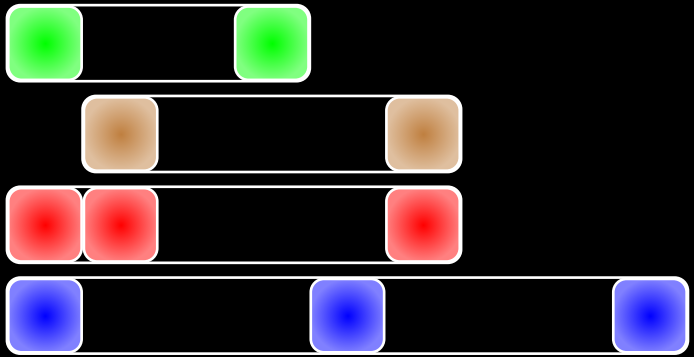

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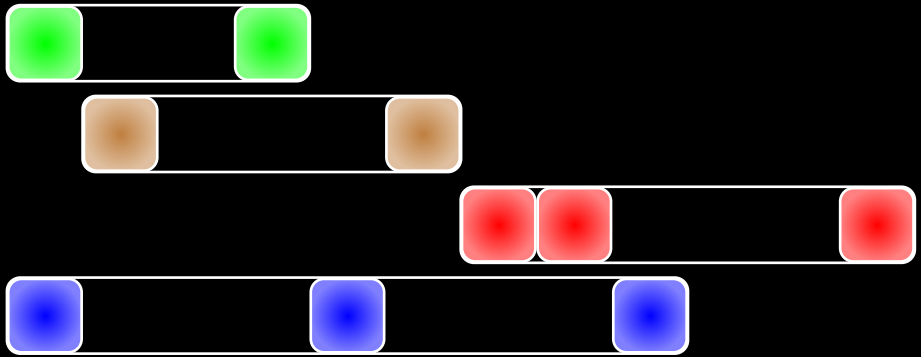

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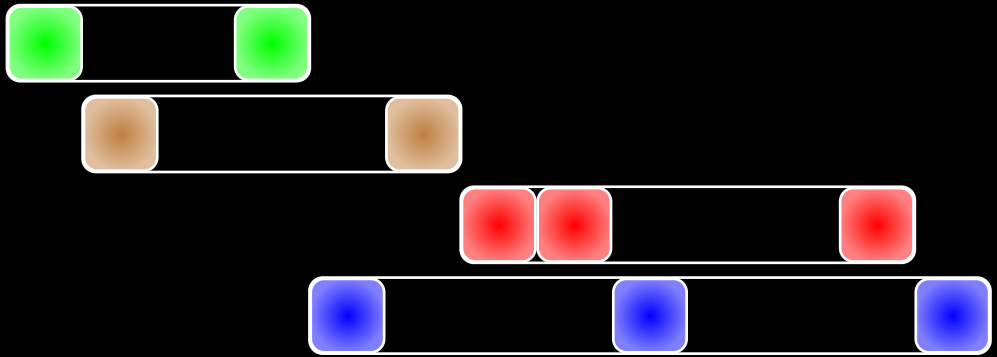

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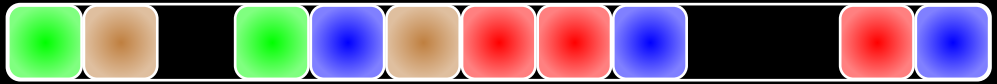


- approximation ratio really so small?
- answer: NO, in fact: $\Theta(\sqrt{m})$, where m is the optimal value! Köppl+'25

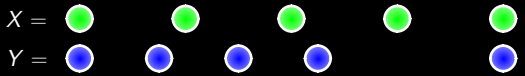
Does it matter?

heuristic allows pre-sorting tiles

- none: no sorting
- incFreq: sort by increasing number of ones
- decFreq: sort by decreasing number of ones
- incDens: sort by increasing density $\frac{\text{number of ones}}{\text{length}}$
- decDens: sort by decreasing density
- random: 10 random shuffling, taking the shortest result

one bad input to rule them all

- only two tile types X and Y
- integer parameters c and g
- $X = (10^g)^c 1$, $Y = (10^{g-1})^c 10^c 1$
- $|X| = |Y| = c(g+1)+1$
- $\#_1 X = \#_1 Y = c+1$
- Request n tiles in alternating order $X, Y, X, Y, \ldots$.



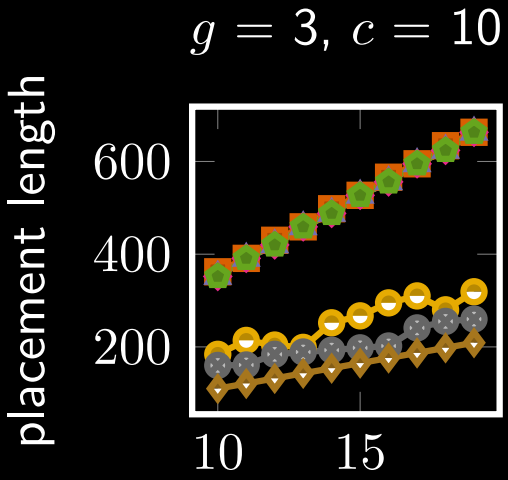
$(c = 5,\ g = 4)$

bad instance 1/2

- lowerbound: total number of ones
- dp: optimal solution

why is random good?

- number of tile types to pick is too small
- best solution for this instance is to sort in order $X, \ldots, X, Y, \ldots, Y$



bad instance 2/2

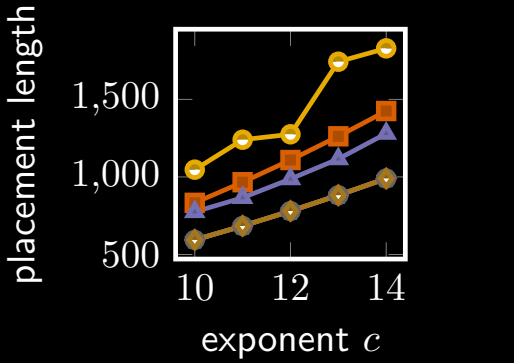
new tile type:

- $Z = (10^{g-2})^c 10^{2c} 1$
- order : $X, Y, Z, X, Y, Z, \ldots$
- add some helper tiles to fill up all holes
  $\Rightarrow \mathrm{dp} = \mathrm{lowerbound}$

random performs worst!

hence: exact algorithms make sense to study

3 tile kinds, $g = 8$, $n = 48$



Parameterized results for minLengthand minMaxShift

	$\mathcal{P}$-time for	$\mathcal{NP}$-hardness even for
ℓ	$\ell \in O(\lg \lg n)$ or [Juge+'26] $\ell \in O(\lg n) \wedge \kappa \in O(\lg n)$	$\ell \in \Omega(\lg n)$ Bannai+'24
ζ	$\zeta = 1$ (trivial)	$\zeta = 2$ [Juge+'26] † $\zeta = 4$ [Juge+'26] ¶
κ	$\kappa \in O(\lg n) \wedge \ell \in O(\lg n)$	$\kappa = 1$ [Juge+'26]
m	$m = \ell$ (trivial) †	$m \leqslant \ell + 2$ Even+'77
ρ	$\rho = 0$ (trivial) ¶	$\rho \leqslant 2$ Even+'77 ¶

- †: concerns only minLength
- ¶: concerns only minMaxShift

- ℓ: maximum length of a tile
- ζ: maximum number of numerals per tile
- κ: number of tile types
- m: optimal placement size
- ρ: maximum shift

from where to reduce? (negative side)

general hardness

- Even+'77: reduction from 3-coloring to MINMAXSHIFT [easy reduction]
- Bannai+'24: reduction from Hamilton path to MINMAXSHIFT and MINLENGTH, mimicking reduction to shortest superstring Gallant+'80 [very technical reduction]
- Kyoto-Group'25: reduction from 3-partition [unpublished, somewhat technical]

hard even for special input

- Juge+'26: reduction from clique [technical] ($\rightarrow$ even one tile type is hard)
- reduction from multi Skolem words Nordh'07 [very easy] ($\rightarrow$ one arbtirary tile and even having at most two 1s in a row is hard)
- Juge+'26: reduction from coupled scheduling problem [very easy] ($\rightarrow$ even having at most two 1s in a row is hard)

algorithmic aspects (positive side)

- easy only for very special input (e.g., constant length)
- otherwise: dynamic programming

dynamic programming

- one tile type: $O(n^2 2^\ell \ell \log \ell)$,
  - ℓ: length of tile
  - fixed-parameter tractable in ℓ!
- general case: $O(n^\kappa \ell n 2^\ell n) \subset O(n^{2^\ell} \ell n 2^\ell n)$ time
- κ: number of tile types

DP for one tile type

naïve approach

- track order explicitly
- gives $\frac{n!}{c_1!c_2!\cdots c_\kappa!}$ possibilities, where
  - c_i is the number of tiles of i-th tile kind
  - exponential in n even for small κ.

solution for one tile type $\kappa = 1$

- track number of taken tiles
- consider only $\Theta(\ell)$ length suffix of all possible solutions

DP table: $D[i][S]$

- i: how many tiles have been used?
- S: ℓ-length suffix
- value: shortest placement length with suffix S using i tiles

DP: general case

general solution: use Parikh vectors

- track count of each taken tile type with vector $\vec{p}$
- $\vec{p} = (p_1, p_2, \ldots, p_\kappa)$ and $p_i \in [0..c_i]$
- do not track the order!

DP table: $D[\vec{p}][S]$

- $\vec{p}$: Parikh vector
  (counts of each taken tile type)
- S: ℓ-length suffix
- value: shortest placement length

Inapproximability

- MINMAXSHIFT: there is a L-reduction from k-coloring, so there is no $\mathcal{P}$-time approximation with ratio $n^{1-\epsilon}$ for some $\epsilon > 0$ unless $\mathcal{NP} = \mathcal{P}$
- MINLENGTH: meta-proof via reduction from any NP-hard problem by adding a gadget: $2 - \epsilon$ approximation not possible unless $\mathcal{NP} = \mathcal{P}$

There is a big gap between both results.

recall

Ziegler's approximation algorithm has an approximation of

- $\Theta(\sqrt{m})$ for MINLENGTH, where m is the length of the optimal output
- $\Theta(\sqrt{\rho})$ for MINMAXSHIFT, where ρ is the allowed maximal shift of the optimal output

Problem

What is the best approximation we can achieve for MINLENGTH?